

Which Test Answers the Question Asked?

Unit 3 · Topic 3.5 · TODAY'S PROBLEM

A district of 2,400 students had bus ridership $p_0 = 0.40$ before a new route. Before collecting data, the director asks whether the current proportion p has *increased*. An SRS of 180 current students finds 84 riders.

Current SRS counts

Group	Bus	Other	Total
Students	84	96	180

Maya: “Use $H_0 : p = 0.40$ versus $H_a : p > 0.40$ and a one-sample z -test. Check large counts with 72 and 108.”

Theo: “Use $H_a : p \neq 0.40$ because samples can differ either way. Check large counts with the observed 84 and 96.”

FIRST TAKE (5 min, on your sheet)

Which setup better answers whether ridership increased? Choose Maya or Theo and cite one detail from the question or sample.

- DOK 1** (a) Define the population parameter p in context and identify the null value p_0 .
- DOK 2** (b) Compute $\hat{p}$, name the testing procedure, and verify the random, 10 percent, and large-counts conditions using the appropriate values.
- DOK 3** (c) Choose the warranted setup. Use the word *increased*, distinguish p_0 from $\hat{p}$, and explain why large counts uses null-expected values. Give the reader consequence of Theo's alternative and count check.

Video + follow-along: u6_lesson4_live.html (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

